

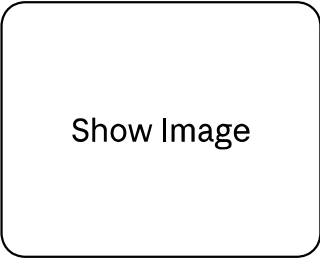
Virtual Try-On

A local AI-powered virtual try-on tool that lets you see yourself wearing any garment from a product image.

What it does

1. Upload any product image of a model wearing a shirt
2. Select your body type, height, and skin tone
3. Scan your face via webcam
4. See yourself wearing that shirt on your chosen body type

Demo



Show Image

Tech Stack

- **Python** — core language
- **OpenCV** — image processing and face detection
- **InsightFace** — realistic AI face swapping
- **MediaPipe** — body pose detection
- **Flask** — web UI (coming soon)

Setup

1. Clone the repo

```
bash
git clone https://github.com/yourusername/virtual-tryon
cd virtual-tryon
```

2. Create virtual environment

```
bash
python -m venv venv
venv\Scripts\activate
```

3. Install dependencies

```
bash

pip install opencv-python mediapipe numpy Pillow
pip install torch torchvision --index-url https://download.pytorch.org/whl/cpu
pip install insightface onnxruntime
pip install trimesh flask
```

4. Download required model

Download `inswapper_128.onnx` from:

```
https://huggingface.co/eziuruan/inswapper_128.onnx/resolve/main/inswapper_128.onnx
```

Place it in the root `virtual-tryon/` folder.

Usage

Step 1 — Detect body pose in garment image

```
bash

python step1_pose.py
```

Step 2 — Detect face in garment image

```
bash

python step2_face.py
```

Step 3 — Capture your face via webcam

```
bash

python step3_capture_user_face.py
```

Press **SPACE** when the green box appears on your face.

Step 4 — Swap your face onto the model

```
bash

python step4b_insightface_swap.py
```

Step 5 — Select body type and skin tone

Edit these lines in `step5_body_select.py`:

```
python
```

```
BODY_TYPE = "fat"      # slim / athletic / average / fat
HEIGHT = "medium"     # short / medium / tall
SKIN_TONE = 4          # 1=very light 2=light 3=medium 4=olive 5=brown 6=dark
```

Then run:

```
bash
```

```
python step5_body_select.py
```

Body Types Available

Body Type	Short	Medium	Tall
Slim		soon	
Athletic	soon	soon	soon
Average	soon	soon	soon
Fat	soon		

Skin Tones

Value	Tone
1	Very Light
2	Light
3	Medium
4	Olive
5	Brown
6	Dark

Roadmap

- ☒ Body pose detection
- ☒ Face detection
- ☒ Webcam face capture

- ☒ AI face swap (InsightFace)
- ☒ Body type selection
- ☒ Skin tone adjustment
- ☐ Web UI with Flask
- ☐ Upload any garment image via browser
- ☐ Real-time webcam face scan in browser
- ☐ 3D model viewer

Notes

- Runs fully locally — no internet needed after setup
- No GPU required — works on CPU
- Built as a portfolio project exploring AI, computer vision, and virtual fashion tech

Author

Built by [Your Name] — [your LinkedIn or portfolio link]